SSU Data Dash 2026: Home-Price Valuation Model Comparison

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Video (YouTube, unlisted): https://youtu.be/04JD9xy812Y
Code and prediction files (GitHub):
https://github.com/tianhui876-a11y/SSU-Data-Dash-2026

Summary

We developed two complementary approaches for synthetic Toronto home-price valuation: histogram-based gradient boosting and tuned-history Lasso regression. Both were evaluated separately on the Off-Market track (no listing price) and the List-Aware track (listing price permitted), using monthly walk-forward validation for December 2025–May 2026. Gradient boosting achieved MdAPE of **11.07%** Off-Market and **4.45%** List-Aware; Lasso achieved **11.79%** and **4.40%**, respectively. The stronger model depends on the track; differences on held-out future sales remain unknown.

1 Data and features

The supplied dataset contains 150,000 labeled synthetic transactions from January 2021 through May 2026. The hidden tests contain 3,974 Off-Market homes and 4,064 List-Aware homes. Models use only information present at prediction time; neither uses sale price, sale date, days on market, external after-sale estimates, or identifiers as predictors. The Off-Market models never use listing price.

Gradient-boosted features. Area-range midpoints, bedrooms, washrooms, parking, garage, lot dimensions/area, coordinates and their product, municipality, community, housing type/style, listing month/year and time trend, plus description length and selected condition keywords. List-Aware additionally uses listing price and its logarithm. Categories are ordinal-encoded and missing numeric inputs receive training-set medians.

Lasso features. Interior area and lot dimensions yield log-area and log(1+lot area); location and housing types are encoded into reference-level dummy variables. Additional predictors include missingness flags, a linear listing-time trend and annual sine/cosine terms. List-Aware adds log listing price. Medians, category levels and encodings are learned from fitting data only; remarks are excluded from Lasso.

2 Models

Histogram-based gradient boosting (HGB). Separate `HistGradientBoostingRegressor` models predict log sale price; outputs are exponentiated and clipped to CAD 10,000–100,000,000. Training uses squared log-price loss (*not* direct MdAPE minimization). The implemented configuration uses learning rate 0.06, 280 boosting iterations, 31 maximum leaf nodes, at least 35 observations per leaf, L2 penalty 2.0 and random seed 42. The `features.py`/`models.py` pipeline preserves the track-specific feature restriction.

Tuned-history Lasso. Separate standardized log-linear regressions are fitted with `glmnet` using an L1 penalty (`alpha = 1`) and an unpenalized intercept. Predictions are exponentiated and rounded to CAD. Candidate penalties and history windows are selected through an inner historical validation. In the reported final fits, each track selected a 12-month training window (22,238 sales): penalty 0.00066443 for Off-Market and 0.000019398 for List-Aware; reported nonzero coefficients were 162/180 and 175/181, respectively. Coefficient selection indicates predictive association, not causation.

3 Validation

For both approaches, the outer evaluation proceeds month by month from **December 2025 through May 2026**. Each evaluation month is predicted using only sales completed before its first day; each track contains **10,069** historical validation homes. No random train/test split is used.

HGB refits both track models on all earlier sales for each month. Lasso additionally holds out the latest three earlier months for inner tuning of penalty and historical window, refits on the selected prior history, and then predicts the outer month. Preprocessing is relearned within each fit, which avoids using future labels to compute medians or category levels.

Interpretation caveats. The Lasso protocol applies monthly sale cutoffs rather than strictly checking availability on each home's listing day. The hidden test spans June–September 2026, so its error need not match historical validation. Repeated model redesign using the same six outer months can also make the apparent validation advantage optimistic.

4 Results

Track	Method	MdAPE (%)	MAPE (%)
Off-Market	Starter (10 nearby earlier sales)	20.84	27.43
Off-Market	HGB + log price	**11.07**	14.19
Off-Market	Tuned-history Lasso	11.79	14.94
List-Aware	Starter (listing price)	7.26	8.09
List-Aware	HGB + log price	4.45	5.55
List-Aware	Tuned-history Lasso	**4.40**	5.54

Table 1: Walk-forward validation (Dec. 2025–May 2026). MdAPE is median absolute percentage error; MAPE is mean absolute percentage error. Starter values follow the teammate's reported benchmark table.

HGB has 0.72 percentage points lower Off-Market MdAPE than Lasso; Lasso has 0.05 points lower List-Aware MdAPE. This narrow List-Aware margin is not evidence of a reliable difference on hidden-test homes. HGB's reported bootstrap 95% MdAPE intervals are 10.85–11.36% (Off-Market) and 4.36–4.56% (List-Aware), based on home-level resampling that may understate month-to-month dependence.

5 Where the model fails

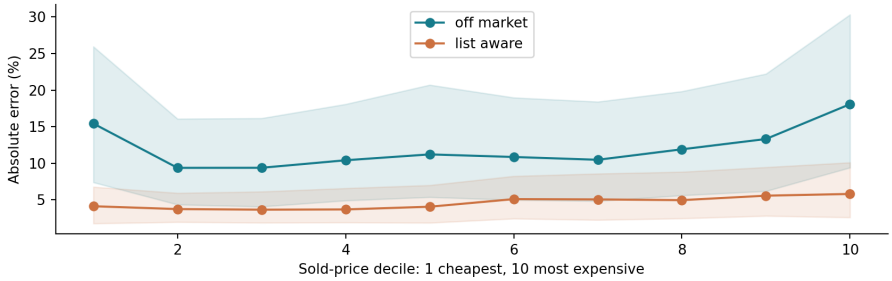


Figure 1: **Lasso only:** validation MdAPE by actual sold-price decile, with the middle 50% of individual absolute percentage errors shaded. Grouping by actual price is retrospective analysis, not a prediction-time feature. Reproduced from the teammate's original Lasso report.

Lasso Off-Market MdAPE is 10.37% for CAD 500,000–1 million homes but 16.41% above CAD 1.5 million. Above CAD 1.5 million, median signed error is −12.58% (underprediction); below CAD 500,000 it is +11.49% (overprediction). Removing geographic inputs increased Off-Market MdAPE by 5.68 percentage points; removing size increased it by 1.91 points. Removing listing price from the List-Aware feature set increased MdAPE by 7.39 points in the teammate's separate Lasso ablation. These are model-specific, retrospective findings; no analogous HGB subgroup results were independently verified.

6 Limitations

The transactions and geography are synthetic, so real-world accuracy is unestablished. Lasso may miss nonlinear interactions; HGB's ordinal encoding may be an imperfect representation of categories. Prices at the tails, market drift and thinly represented property groups remain risks. Lasso's historical empirical residual-band coverage was reported as 89.67% Off-Market and 89.12% List-Aware, with no guaranteed future coverage. Validation performance is not hidden-test performance; the Lasso metrics have not been independently rerun in this report.

Outside data

Source	URL	Periods covered
None	–	–

Reproducing our results

HGB: From `HistGradientBoosting/`, run `pip install -r requirements.txt`; then `python run.py validate`, `python run.py report`, `python run.py predict`, and `python run.py check`.
Lasso: The `Lasso/SSU_Lasso_File_Chooser.R` script needs R, `glmnet`, `Matrix`, and the three official CSV inputs, selected in order (training, Off-Market, List-Aware). It generates model-specific `submissions/` files.
Final submission status: Both sets of model-specific CSVs are in their respective repository subfolders. The team must designate and validate the final pair in the repository-root `submissions/` before submission.

Sources: Organizer starter, rules, and report template: SSU-Data-Dash; data: SSU_data_dash; Lasso method: Tibshirani (1996), Friedman et al. (2010).